[INCH-POUND]
A-A-55564/3
15 February, 1999

COMMERCIAL ITEM DESCRIPTION
SPECIFICATION SHEET

# RESISTOR, VOLTAGE SENSITIVE (VARISTOR), METAL OXIDE, RADIAL LEAD

The General Services Administration has authorized the use of this commercial item description (CID) for all federal agencies.

1. SCOPE. This CID covers the general requirements for a radial leaded, transient voltage suppression device. The varistors covered by this CID are intended for commercial applications.

2. CLASSIFICATION. This CID uses a classification system which is included in the Part Identification Number (PIN) as shown in the following example (see 7.1).

| AA55564 | 03 | -001 |
|---|---|---|
| CID number | Applicable CID sheet (two digit) | Dash number (see applicable CID sheet) |

3. SALIENT CHARACTERISTICS.

3.1 Interface and physical dimensions. The varistors supplied to this CID shall be as specified herein (see figure 1).

3.2 Power rating. The power rating for the varistor shall be 0.20 watt, 0.25 watt, 0.40 watt, 0.60 watt, and 1.00 watt for cases 1, 2, 3, 4 and 5 respectively.

3.3 Voltage rating. The voltage rating shall be as specified in table I.

3.4 Energy rating. The energy rating shall be as specified in table I.

3.5 Peak current. The peak current rating shall be as specified in table I.

3.6 Clamping voltage. The clamping voltage shall be as specified in table I.

3.7 Capacitance. The capacitance shall be as specified in table I.

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any data which may improve this document should be sent to: Defense Supply Center, Columbus, ATTN: DSCC VAM, 3990 East Broad Street, Columbus OH 43213-1199, or telephone (614) 692-0552, or facsimile (FAX) (614) 692-1644.

AMSC N/A FSC 5905

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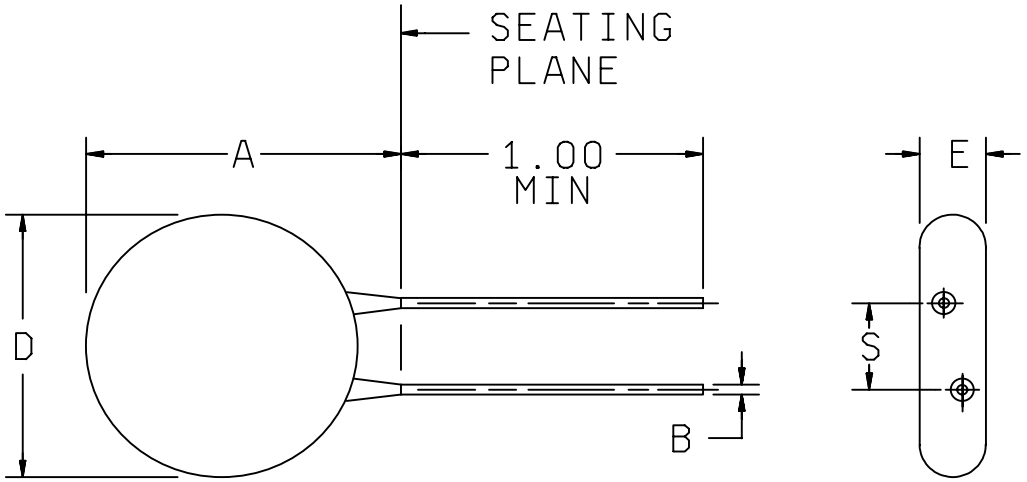


| Case code | Dimensions | | | | |
|---|---|---|---|---|---|
| | A (max) | D (max) | E (max) | S ±0.039 | B (max) |
| 1 | 0.394 | 0.296 | 0.220 | 0.199 | 0.227 |
| 2 | 0.472 | 0.354 | 0.220 | 0.199 | 0.227 |
| 3 | 0.630 | 0.492 | 0.220 | 0.296 | 0.034 |
| 4 | 0.787 | 0.689 | 0.220 | 0.296 | 0.034 |
| 5 | 1.043 | 0.906 | 0.220 | 0.296 | 0.034 |

| Inch | mm | Inch | mm | Inch | mm | Inch | mm | Inch | mm |
|---|---|---|---|---|---|---|---|---|---|
| 0.027 | 0.69 | 0.197 | 5.00 | 0.354 | 8.99 | 0.492 | 12.50 | 0.787 | 19.99 |
| 0.034 | 0.86 | 0.220 | 5.59 | 0.394 | 10.01 | 0.630 | 16.00 | 1.000 | 25.40 |
| 0.039 | 0.99 | 0.296 | 7.52 | 0.472 | 11.99 | 0.689 | 17.50 | 1.043 | 26.49 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 1. Case dimensions and configuration.



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TABLE I. Rating and characteristics.

| Dash number | Case code | Voltage rating (max) | | Energy in joules (max) | Peak current | Varistor voltage with 1mA dc test current | | Max clamping voltage at test current | | Typical capacitance |
|---|---|---|---|---|---|---|---|---|---|---|
| | | (rms) | (dc) | (J) | (amperes) | (min) | (max) | (V) | (I) | picofarads |
| 001 | 1 | 14 | 18 | 0.20 | 35 | 18.7 | 26 | 51 | 2 | 400 |
| 002 | 2 | 14 | 18 | 0.90 | 150 | 18.7 | 26 | 47 | 5 | 1600 |
| 003 | 3 | 14 | 18 | 2.00 | 350 | 18.7 | 26 | 43 | 5 | 400 |
| 004 | 4 | 14 | 18 | 4.00 | 750 | 18.7 | 26 | 43 | 10 | 900 |
| 005 | 5 | 14 | 18 | 6.50 | 1500 | 19.2 | 26 | 43 | 20 | 18000 |
| 006 | 1 | 17 | 22 | 0.25 | 35 | 23 | 31.1 | 59 | 2 | 300 |
| 007 | 2 | 17 | 22 | 1.00 | 150 | 23 | 31.1 | 57 | 5 | 1300 |
| 008 | 3 | 17 | 22 | 2.50 | 350 | 23 | 31.1 | 53 | 5 | 3000 |
| 009 | 4 | 17 | 22 | 5.00 | 750 | 23 | 31.1 | 53 | 10 | 7000 |
| 010 | 5 | 17 | 22 | 8.00 | 1500 | 23 | 31.1 | 50 | 20 | 15000 |
| 011 | 1 | 20 | 26 | 0.30 | 35 | 29.5 | 38 | 67 | 2 | 250 |
| 012 | 2 | 20 | 26 | 1.20 | 150 | 29.5 | 36.5 | 68 | 5 | 1100 |
| 013 | 3 | 20 | 26 | 3.00 | 350 | 29.5 | 36.5 | 64 | 5 | 2700 |
| 014 | 4 | 20 | 26 | 6.00 | 750 | 29.5 | 36.5 | 64 | 10 | 6000 |
| 015 | 5 | 21 | 27 | 9.00 | 1500 | 29.5 | 36.5 | 58 | 20 | 13000 |
| 016 | 5 | 23 | 31 | 10.00 | 1500 | 32 | 40 | 63 | 20 | 12000 |
| 017 | 1 | 25 | 31 | 0.35 | 35 | 35 | 46 | 79 | 2 | 220 |
| 018 | 2 | 25 | 31 | 1.50 | 150 | 35 | 43 | 79 | 5 | 900 |
| 019 | 3 | 25 | 31 | 3.50 | 350 | 35 | 43 | 76 | 5 | 2200 |
| 020 | 4 | 25 | 31 | 7.20 | 750 | 35 | 43 | 76 | 10 | 5000 |
| 021 | 1 | 30 | 38 | 0.40 | 35 | 42 | 55 | 90 | 2 | 200 |
| 022 | 2 | 30 | 38 | 1.80 | 150 | 42 | 52 | 92 | 5 | 800 |
| 023 | 3 | 30 | 38 | 4.50 | 350 | 42 | 52 | 89 | 5 | 2000 |
| 024 | 4 | 30 | 38 | 8.80 | 750 | 42 | 52 | 89 | 10 | 4500 |
| 025 | 1 | 35 | 45 | 0.50 | 35 | 50 | 66 | 108 | 2 | 180 |
| 026 | 2 | 35 | 45 | 2.30 | 150 | 50 | 62 | 107 | 5 | 700 |
| 027 | 3 | 35 | 45 | 5.50 | 350 | 50 | 62 | 103 | 5 | 1800 |
| 028 | 4 | 35 | 45 | 10.00 | 750 | 50 | 62 | 103 | 110 | 3900 |
| 029 | 1 | 40 | 56 | 0.60 | 35 | 61 | 80 | 127 | 2 | 150 |
| 030 | 2 | 40 | 56 | 3.00 | 150 | 61 | 75 | 127 | 5 | 600 |
| 031 | 3 | 40 | 56 | 6.50 | 350 | 61 | 75 | 123 | 5 | 1500 |
| 032 | 4 | 40 | 56 | 13.00 | 750 | 61 | 75 | 123 | 10 | 3300 |
| 033 | 1 | 50 | 66 | 1.20 | 150 | 73 | 97 | 145 | 2 | 120 |
| 034 | 2 | 50 | 66 | 3.50 | 500 | 73 | 91 | 135 | 10 | 500 |
| 035 | 3 | 50 | 66 | 7.30 | 1000 | 73 | 91 | 135 | 25 | 1100 |
| 036 | 4 | 50 | 66 | 13.00 | 2000 | 73 | 91 | 145 | 50 | 2500 |
| 037 | 1 | 60 | 81 | 1.50 | 150 | 90 | 117 | 175 | 2 | 90 |
| 038 | 2 | 60 | 81 | 4.30 | 500 | 90 | 110 | 165 | 10 | 400 |
| 039 | 3 | 60 | 81 | 8.90 | 1000 | 90 | 110 | 165 | 25 | 900 |



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TABLE I Rating and characteristics – Continued.

| Dash number | Case code | Voltage rating (max) | | Energy in joules (max) | Peak current | Varistor voltage with 1mA dc test current | | Max clamping voltage at test current | | Typical capacitance |
|---|---|---|---|---|---|---|---|---|---|---|
| | | (rms) | (dc) | (J) | (amperes) | (min) | (max) | (V) | (I) | picofarads |
| 040 | 4 | 60 | 81 | 16.00 | 2000 | 90 | 110 | 175 | 50 | 2000 |
| 041 | 1 | 75 | 102 | 1.80 | 150 | 108 | 138 | 205 | 2 | 70 |
| 042 | 2 | 75 | 102 | 5.30 | 500 | 108 | 132 | 205 | 10 | 300 |
| 043 | 3 | 75 | 102 | 11.00 | 1000 | 108 | 132 | 200 | 25 | 750 |
| 044 | 4 | 75 | 102 | 19.00 | 2000 | 108 | 132 | 210 | 50 | 1700 |
| 045 | 1 | 92 | 127 | 2.30 | 200 | 135 | 173 | 240 | 2 | 60 |
| 046 | 2 | 95 | 127 | 6.50 | 750 | 135 | 165 | 250 | 10 | 250 |
| 047 | 3 | 95 | 127 | 13.00 | 1500 | 135 | 165 | 250 | 25 | 600 |
| 048 | 4 | 95 | 127 | 23.00 | 3000 | 135 | 165 | 255 | 50 | 1400 |
| 049 | 1 | 115 | 153 | 2.70 | 200 | 162 | 207 | 290 | 2 | 50 |
| 050 | 2 | 115 | 153 | 7.70 | 750 | 162 | 198 | 295 | 10 | 200 |
| 051 | 3 | 115 | 153 | 16.00 | 1500 | 162 | 198 | 300 | 25 | 500 |
| 052 | 4 | 115 | 153 | 27.00 | 3000 | 162 | 198 | 300 | 50 | 1100 |
| 053 | 1 | 4 | 5.5 | 0.10 | 50 | 6 | 11 | 30 | 1 | 1400 |
| 054 | 2 | 4 | 5.5 | 0.40 | 100 | 6 | 11 | 22 | 2.5 | 3000 |
| 055 | 3 | 4 | 5.5 | 0.80 | 250 | 6 | 11 | 20 | 5 | 7500 |
| 056 | 1 | 6 | 8 | 0.14 | 50 | 9 | 16 | 37 | 1 | 1200 |
| 057 | 2 | 6 | 8 | 0.60 | 100 | 9 | 16 | 34 | 2.5 | 2500 |
| 058 | 3 | 6 | 8 | 1.20 | 250 | 9 | 16 | 30 | 5 | 6000 |
| 059 | 1 | 10 | 14 | 0.17 | 100 | 14.4 | 21.6 | 36 | 1 | 1000 |
| 060 | 2 | 10 | 14 | 0.80 | 250 | 14.4 | 21.6 | 36 | 2.5 | 2000 |
| 061 | 3 | 10 | 14 | 1.50 | 500 | 14.4 | 21.6 | 36 | 5 | 5000 |
| 062 | 4 | 10 | 14 | 3.50 | 1000 | 14.4 | 21.6 | 36 | 10 | 11000 |
| 063 | 5 | 10 | 14 | 80.00 | 2000 | 14.4 | 21.6 | 37 | 20 | 22000 |
| 064 | 5 | 25 | 31 | 20.00 | 2000 | 35 | 43 | 76 | 20 | 12000 |
| 065 | 5 | 30 | 38 | 23.00 | 2000 | 42 | 52 | 93 | 20 | 11000 |
| 066 | 5 | 35 | 45 | 30.00 | 2000 | 50 | 62 | 110 | 20 | 10000 |
| 067 | 5 | 40 | 56 | 33.00 | 2000 | 61 | 75 | 135 | 20 | 10000 |
| 068 | 5 | 75 | 2 | 33.00 | 6500 | 108 | 132 | 210 | 100 | 1500 |
| 069 | 5 | 95 | 127 | 45.00 | 6500 | 135 | 135 | 250 | 100 | 1000 |
| 070 | 5 | 115 | 153 | 52.00 | 6500 | 162 | 198 | 300 | 100 | 2400 |
| 071 | 1 | 130 | 170 | 5.50 | 400 | 184 | 226 | 340 | 5 | 100 |
| 072 | 1 | 140 | 180 | 6.00 | 400 | 198 | 253 | 360 | 5 | 90 |
| 073 | 1 | 150 | 200 | 7.00 | 400 | 216 | 264 | 395 | 5 | 80 |
| 074 | 1 | 175 | 225 | 7.50 | 400 | 243 | 311 | 455 | 5 | 70 |
| 075 | 1 | 210 | 275 | 9.00 | 400 | 297 | 380 | 540 | 5 | 60 |
| 076 | 1 | 230 | 300 | 9.50 | 400 | 324 | 396 | 595 | 5 | 55 |
| 077 | 1 | 250 | 330 | 10.00 | 400 | 351 | 449 | 650 | 5 | 50 |
| 078 | 1 | 275 | 369 | 11.00 | 400 | 387 | 495 | 710 | 5 | 45 |
| 079 | 1 | 300 | 385 | 12.00 | 400 | 420 | 517 | 775 | 5 | 35 |
| 080 | 1 | 385 | 505 | 13.00 | 400 | 558 | 682 | 1025 | 5 | 33 |
| 081 | 1 | 420 | 560 | 14.00 | 400 | 610 | 748 | 1120 | 5 | 32 |
| 082 | 1 | 440 | 585 | 15.50 | 400 | 643 | 787 | 1180 | 5 | 31 |
| 083 | 1 | 460 | 615 | 17.00 | 400 | 675 | 825 | 1240 | 5 | 30 |

3.8 Operating ambient temperature. The ambient temperature shall be - 55°C to + 85°C.

3.9 Storage temperature. The storage temperature shall be - 55°C to + 125°C.

3.10 Marking. The varistors supplied to this CID shall be marked with the manufacturer's standard commercial PIN.



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4. REGULATORY REQUIREMENTS. This section is not applicable to this CID sheet.

5. QUALITY ASSURANCE PROVISIONS. Quality assurance provisions shall be as in A-A-55564.

6. PACKAGING. Packaging shall be as specified in A-A-55564.

7. NOTES.

7.1 PIN. The PIN should be used for Government purposes to buy commercial products to this CID. See section 2 for PIN format example.

7.2 Commercial and Government Entity (CAGE) code. For ordering purposes, inventory control, and submission of these varistors to DSCC under the Military Parts Control Advisory Group (MPCAG) evaluation program, CAGE code 58536 should be used.

7.3 Source of documents.

Commercial Item Description

A-A-55564 - Resistor, Voltage Sensitive (Varistor), General Requirements for.

(Copies of commercial item descriptions are available from the Defense Automated Printing Service, Building 4D (DPM-DoDSSP), 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

7.4 Ordering data. Ordering data shall be as specified in A-A-55564.

7.5 Commercial products. As part of the marketing analysis and research effort, this CID was coordinated with the following manufacturers of commercial products. At the time of CID preparation and coordination, these manufacturers were know to have commercial products that would meet the requirements of this CID. (NOTE: This information should not be considered as a list of approved manufacturers or be used to restrict procurement to only the manufacturers shown).

| MFGR's CAGE | MFGR's name and address |
|---|---|
| S6019 | Harris Corp.<br>Harris Semiconductor Division<br>724 Route 202<br>P.O. Box 591<br>Somerville, NY 08876-0591 |
| | Manufactured by<br>Harris Corp. - ECCO, Ltd.<br>The Demesne<br>Dundalk, county Loutch<br>Republic of Ireland |



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7.6 Part number (P/N) supersession data. This CID supersedes the following manufacturer's P/N's as shown in table II. This information is being provided to assist in reducing proliferation in the Government inventory system.

TABLE II. P/N supersession data.

| Dash number (see table I) AA556403- | Manufacturer's P/N | Dash number (see table I) AA556403- | Manufacturer's P/N | Dash number (see table I) AA556403- | Manufacturer's P/N |
|---|---|---|---|---|---|
| -001 | V22ZA05 | -029 | V68ZA05 | -057 | V12ZA1 |
| -002 | V22ZA1 | -030 | V68ZA2 | -058 | V12ZA2 |
| -003 | V22ZA2 | -031 | V68ZA3 | -059 | V18ZA05 |
| -004 | V22ZA3 | -032 | V68ZA10 | -060 | V18ZA1 |
| -005 | V24ZA50 | -033 | V82ZA05 | -061 | V18ZA2 |
| -006 | V27ZA05 | -034 | V82ZA2 | -062 | V18ZA3 |
| -007 | V27ZA1 | -035 | V82ZA4 | -063 | V18ZA40 |
| -008 | V27ZA2 | -036 | V82ZA12 | -064 | V39ZA20 |
| -009 | V27ZA4 | -037 | V100ZA05 | -065 | V47ZA20 |
| -010 | V27ZA60 | -038 | V100ZA3 | -066 | V56ZA20 |
| -011 | V33ZA05 | -039 | V100ZA4 | -067 | V68ZA20 |
| -012 | V33ZA1 | -040 | V100ZA15 | -068 | V120ZA20 |
| -013 | V33ZA2 | -041 | V120ZA05 | -069 | V150ZA20 |
| -014 | V33ZA5 | -042 | V120ZA1 | -070 | V180ZA20 |
| -015 | V33ZA70 | -043 | V120ZA4 | -071 | V205ZA05 |
| -016 | V36ZA80 | -044 | V120ZA6 | -072 | V220ZA05 |
| -017 | V39ZA05 | -045 | V150ZA05 | -073 | V204ZA05 |
| -018 | V39ZA1 | -046 | V150ZA1 | -074 | V270ZA05 |
| -019 | V39ZA3 | -047 | V150ZA4 | -075 | V330ZA05 |
| -020 | V39ZA6 | -048 | V150ZA8 | -076 | V360ZA05 |
| -021 | V47ZA05 | -049 | V180ZA05 | -077 | V390ZA05 |
| -022 | V47ZA1 | -050 | V180ZA1 | -078 | V430ZA05 |
| -023 | V47ZA3 | -051 | V180ZA5 | -079 | V470ZA05 |
| -024 | V47ZA7 | -052 | V180ZA10 | -080 | V620ZA05 |
| -025 | V56ZA05 | -053 | V8ZA05 | -081 | V680ZA05 |
| -026 | V56ZA2 | -054 | V8ZA1 | -082 | V715ZA05 |
| -027 | V56ZA3 | -055 | V8ZA2 | -083 | V750ZA05 |
| -028 | V56ZA8 | -056 | V12ZA05 | | |

7.6 Government users. To acquire information on obtaining these varistors from the Government inventory system, contact Defense Supply Center, Columbus, ATTN: DSCC-CPB, 3990 East Broad Street, Columbus, OH 43213-5000, or telephone (614) 692-7678.

MILITARY INTERESTS:

Custodian:

Navy - EC

CIVIL AGENCY COORDINATING ACTIVITY

GSA - 7FXE

Preparing Activity:

DLA-CC

Project 5905-1542



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